- "**JFB = Jacobian-Free Backpropagation**, replaces that inverse with just $\partial T/\partial \theta$. One step instead of a full implicit solve."
- "Combined with $B_k \approx \nabla_u f$, we get the **central training formula**."
- "This is what we actually optimize."

2.8: Algorithm summary (1min)

1. "Sample initial state from ρ."
2. "Roll out in the environment."
3. "Compute controls via the implicit fixed point."
4. "Update Jacobian estimates A_k, B_k."
5. "Backward adjoint pass."
6. "JFB parameter update."

> *if no more time, jump here and skip 2.10–2.13*

2.9 & 2.10: Benchmark problems (45sec)

Van der Pol (2.16):

- "Nonlinear oscillator, scalar control u acts as an external force, goal is to drive (x_1, x_2) to the origin."
- "Chose 3 problems with increasing difficulty (diff control penalty + control coupling) to see wether our model can adapt well to more complex settings"
- "first is explicit (hamiltonian can be solved for u), and the two others are implicit"

Portfolio (2.17):

- "Wealth dynamics with risky and risk-free assets. The drift μ and rate r are **unknown**, exactly the RL setting."
- "Trade-off: maximize log terminal wealth, penalize aggressive positions via $\exp(\pi^2)$."

2.11 to 2.214: Results (1min)

Van der Pol (2.21):

- "The RL-trained controller matches the **oracle** (which has full access to f) in all three problems. Convergence to the origin with no dynamics model."
- "Left: evolution of the loss, Right: convergence to the origin"
- "Autodiff method is slightly better, but our model could potentially outperform it the dimension was higher."

Portfolio (2.22):

- "Left: training loss decreasing smoothly."
- "Right: learned policy rollout, wealth trajectory."
- "Mid-training snapshot at epoch 200 shows policy is already shaping correctly."
- **Analytical wealth:** the true wealth is almost aligned on ours.
- **Analytical policy:** the true policy ($\mu=0.10$, $r=0.03$) has a constant value of 0.85.

2.15: Practical challenges (45 sec)

- "**Locality**: RLS estimates become stale if rollouts drift."
- "**Exploration**: we need enough perturbation σ to identify B_k."
- "**Long-horizon sensitivity**: A_k errors compound through the backward adjoint."

2.16: Outlook (30sec)

- "Same Gelphman pipeline: $\phi\theta \to \nabla\phi\theta \to u^*_\theta \to J(\theta)$."
- "The only structural change: f and ∇f are replaced by local estimates A_k, B_k."
- "Next: unknown running cost L, stochastic dynamics, formal descent guarantees, multi-asset portfolios."
